Gandaki University

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Bachelor of Information Technology

BSM 102

Exercise on Infinite Series

1. Find the sum of the series $\sum_{n=0}^{\infty} x^n$, where $|x| < 1$

2. Consider the series $\sum\limits_{n=0}^{\infty} \dfrac{n}{(n+1)!}$
   (a) Find the partial sums s_1, s_2, s_3, s_4, Do you recognize the denominators? Use the patterns to guess a formuls for s_n
   b) Show that the given infinite series is convergent and find its sum.

3. Determine whether the series is covergent or divergent. If it is convergent, find its sum.
   (a) $\sum_{n=1}^{\infty} \dfrac{3^n + 2^n}{6^n}$
   (b) $\sum_{n=1}^{\infty} \dfrac{1}{2^{n-1}} - \dfrac{2}{3^{n-1}}$
   (c) $\sum_{n=1}^{\infty} 3^{-n}8^{n+1}$
   (d) $\sum_{n=1}^{\infty} \dfrac{-3^{n-1}}{4^n}$
   (e) $\sum_{n=1}^{\infty} \dfrac{1}{4n^2 - 1}$
   (f) $\sum_{n=1}^{\infty} \dfrac{1}{e^{2n}}$

4. Determine if the following series is convergent or divergent.
   a) $\sum_{n=0}^{\infty} \dfrac{n+2}{2n+7}$ b) $\sum_{n=0}^{\infty} \dfrac{(-1)^n}{n^2+1}$ c) $\sum_{n=1}^{\infty} \dfrac{9^n}{(-2)^{n+1}n}$
   d) $\sum_{n=0}^{\infty} \dfrac{(\pi + \pi i)^{2n+1}}{(2n+1)!}$ e) $\sum_{n=1}^{\infty} \dfrac{1}{\sqrt{2n}}$ f) $\sum_{n=0}^{\infty} \dfrac{(-1)^n(1+i)^{2n}}{(2n)!}$
   g) $\sum_{n=1}^{\infty} \dfrac{(3i)^n n!}{n^n}$ h) $\sum_{n=1}^{\infty} \dfrac{(-1)^n}{n}$ i) $\sum_{n=0}^{\infty} \dfrac{n+i}{3n^2+2i}$
   j) $\sum_{n=0}^{\infty} \dfrac{(20+30i)^n}{n!}$ k) $\sum_{n=2}^{\infty} \dfrac{(-i)^n}{\ln n}$ l) $\sum_{n=1}^{\infty} n^2 \left(\dfrac{1}{4}\right)^n$
   m) $\sum_{n=0}^{\infty} \dfrac{i^n}{n^2 - i}$ n) $\sum_{n=1}^{\infty} \left(\dfrac{n-1}{2n+3}\right)^n$ o) $\sum_{n=1}^{\infty} \left(\dfrac{2n^2-1}{n^2+3}\right)^n$
   p) $\sum_{n=1}^{\infty} \dfrac{(\ln n)^{2n}}{n^n}$ q) $\sum_{n=1}^{\infty} \dfrac{n}{2^n}$

5. Find the center, radius and interval of convergence of the power series

   (a) $\sum_{n=1}^{\infty} 2^n (z-1)^n$

   (b) $\sum_{n=1}^{\infty} \dfrac{(z-2i)^n}{5^n}$

   (c) $\sum_{n=1}^{\infty} \dfrac{(z-2i)^n}{n^n}$

   (d) $\sum_{n=1}^{\infty} \dfrac{(-1)^n n}{8^n} z^n$

   (e) $\sum_{n=1}^{\infty} \dfrac{n^n}{n!}(z-\pi i)^n$

   (f) $\sum_{n=1}^{\infty} \dfrac{(z-2i)^n}{n^n}$

   (g) $\sum_{n=0}^{\infty} \dfrac{n+1}{(2n+1)!}(z-2)^n$

   (h) $\sum_{n=0}^{\infty} \dfrac{n(n-1)}{2^n}(z+i)^{2n}$

6. The series for $\sin x$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \frac{x^{11}}{11!} + \cdots$$

converges to $\sin x$ for all $x \in \mathbb{R}$.

   (a) Find the first six nonzero terms of the series for $\cos x$. For what values of x does the series converge?

   (b) By replacing x with $2x$ in the series for $\sin x$, find a power series that converges to $\sin(2x)$ for all x.

   (c) Using your result from part (a) and series multiplication, calculate the first six nonzero terms of the power series for $2 \sin x \cos x$. Compare your result with your answer from part (b).

7. The series for e^x

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \cdots$$

converges to e^x for all $x \in \mathbb{R}$.

   (a) Find a power series for $\frac{d}{dx}e^x$. Do you get the same series? Explain your answer.

   (b) Find a power series for $\int e^x\,dx$. Do you get the same series (plus a constant)? Explain.

   (c) Replace x with $-x$ in the series for e^x to find a power series that converges to e^{-x}. Then, multiply the series for e^x and e^{-x}, and find the first six nonzero terms of the resulting power series. What function does this new series represent?

8. The series for $\tan x$

$$\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \frac{17x^7}{315} + \frac{62x^9}{2835} + \cdots$$

converges to $\tan x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$.

   (a) Find the first five nonzero terms of the power series for $\ln|\sec x|$. For what values of x does the series converge?